

Operating Systems for Mobiles



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An operating system (OS) is a collection of programs that control the application software that users run and provide a link between the hardware and software currently running on the computer. The OS is also responsible for management and control of all resources such as memory, hard drive and monitor that are shared among the different application programs that may be running simultaneously.

A mobile OS is an OS for devices such as smartphones, tablets, PDAs and other mobile devices.

ios

ios (originally iPhone OS) is a mobile OS developed by Apple Inc. and distributed exclusively for Apple hardware. It was unveiled with the iPhone at Macworld Con-

ference & Expo on January 9, 2007, and released in June of that year.

Architecture

The iOS software development kit (SDK) contains the tools and interfaces needed to develop, install, run and test native apps that appear on an iOS device's Home screen. Native apps are built using iOS system frameworks and Objective-C language and run directly on iOS. Unlike Web apps, native apps are installed physically on a device and are therefore always available to the user, even when the device is in Airplane mode. These reside next to other system apps, and both the app and any user data is synced to the user's computer through iTunes.

At the highest level, iOS acts as an intermediary between the underlying hardware and the apps you create. Apps do not talk to the underlying hardware directly. Instead, these communicate with the hardware through a set of well-defined system interfaces. These interfaces make it easy to write apps that work consistently on devices having different hardware capabilities.

The implementation of iOS technologies can be viewed as a set of layers, which are shown in Fig. 3. Lower layers contain fundamental services and technologies. Higher-level layers build upon the lower layers and provide more sophisticated services and technologies.

Cocoa Touch layer. This layer contains key frameworks for building iOS apps.



Fig. 1: Interconnection between user, application, OS and hardware



Fig. 2: iOS 8 Home screen

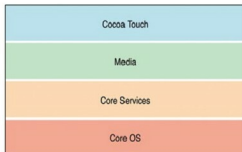


Fig. 3: Layers of iOS architecture